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# Homework 3: DCGAN on WikiArt with FID/IS Evaluation

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## 1 Task, Data, and Model

I train a Deep Convolutional GAN (DCGAN) (1) to generate  $64 \times 64$  RGB artworks from WikiArt. I report (i) training configuration and loss curves, (ii) quantitative evaluation using Fréchet Inception Distance (FID) (2) and Inception Score (IS), and (iii) qualitative sample grids across checkpoints.

**Data.** I use the WikiArt training split. Images are resized to  $64 \times 64$  and normalized to match the GAN input pipeline. Training uses mini-batches of size 128.

**Model and objective.** The generator  $G(z)$  maps  $z \sim \mathcal{N}(0, I)$  to an RGB image and the discriminator  $D(x)$  predicts real/fake. I use the standard non-saturating GAN objective with Adam optimization.

**Evaluation protocol.** For each checkpoint, I generate  $N = 5000$  samples (effective  $N = 4992$  due to batching) and compute FID and IS using an Inception-v3 backbone evaluated at  $299 \times 299$  (3). I treat FID as the primary metric, while IS is reported for completeness.

## 2 Hyperparameters and Experimental Design

**Common.** Image size 64; training batch size 128; Adam learning rate  $2 \times 10^{-4}$ ;  $\beta_1 = 0.5$ ; latent dimension  $z_{\text{dim}} = 128$ . Evaluation uses batch size 64,  $N = 4992$ , and IS is computed with 10 splits. All runs use Apple MPS.

**Part 1 (baseline).** Train from scratch for 30 epochs; evaluate epochs 20/25/30.

**Part 2 (resume).** Resume from the epoch-30 checkpoint and continue training to epoch 80 without re-initialization; evaluate epochs 40/50/60/70/80.

## 3 Results

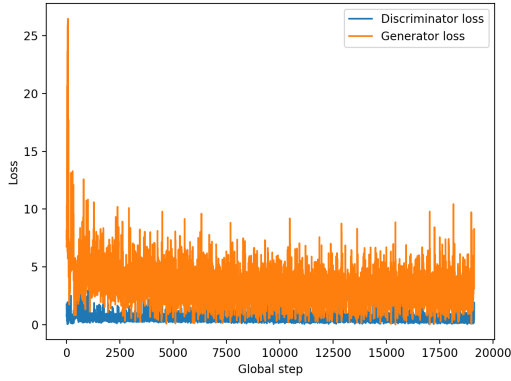
### 3.1 Training Dynamics and Part 1 Metrics

Figure 1 and Table 1 summarize training dynamics and Part 1 quantitative results. Among baseline checkpoints, epoch 25 achieves the lowest FID.

### 3.2 Part 1: Qualitative Samples (20/25/30)

### 3.3 Part 2: Resume Training (40/50/60/70/80)

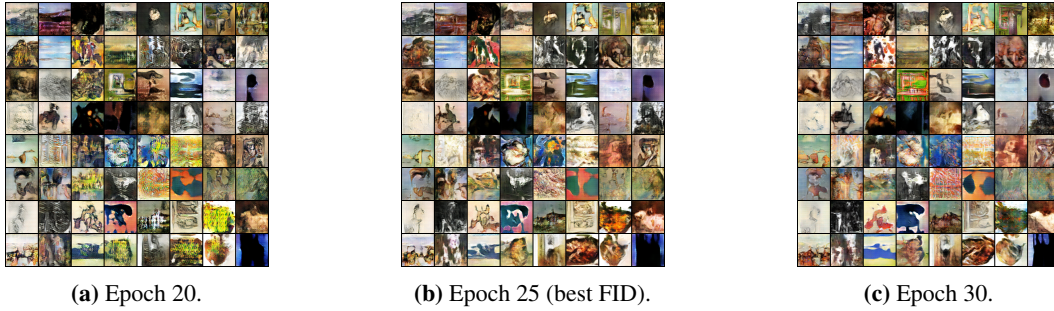
Continuing training substantially improves FID. Table 2 shows a consistent downward trend in FID, with the best checkpoint at epoch 80.



**Figure 1:** DCGAN training losses on WikiArt.

| Checkpoint      | FID ↓          | IS ↑                | #Gen |
|-----------------|----------------|---------------------|------|
| Epoch 20        | 64.2577        | $4.2617 \pm 0.1728$ | 4992 |
| <b>Epoch 25</b> | <b>62.3171</b> | $4.4912 \pm 0.2229$ | 4992 |
| Epoch 30        | 65.7466        | $4.9225 \pm 0.2213$ | 4992 |

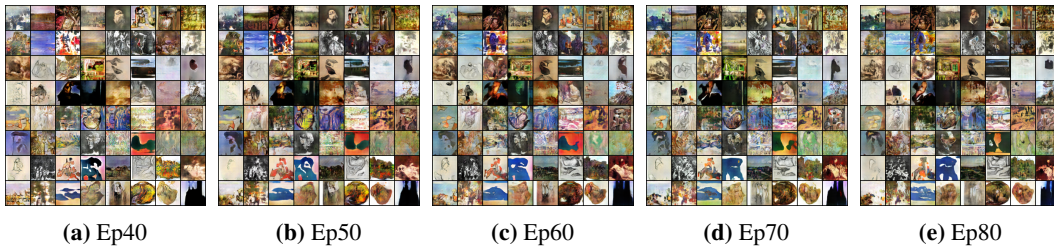
**Table 1:** Part 1 FID/IS on WikiArt (mean  $\pm$  std over 10 splits).



**Figure 2:** Part 1 qualitative sample grids.

| Checkpoint             | FID ↓          | IS ↑                | #Gen |
|------------------------|----------------|---------------------|------|
| Epoch 40               | 53.6920        | $4.9249 \pm 0.1655$ | 4992 |
| Epoch 50               | 47.9007        | $4.8317 \pm 0.1206$ | 4992 |
| Epoch 60               | 46.2842        | $5.0800 \pm 0.1768$ | 4992 |
| Epoch 70               | 46.0689        | $4.7850 \pm 0.0881$ | 4992 |
| <b>Epoch 80 (best)</b> | <b>44.6105</b> | $5.0359 \pm 0.2130$ | 4992 |

**Table 2:** Part 2 (resume) FID/IS on WikiArt (mean  $\pm$  std over 10 splits).



**Figure 3:** Part 2 qualitative sample grids (resume training).

## 4 Discussion

**Effect of longer training.** The best Part 1 checkpoint (epoch 25) achieves FID 62.3171, while the best Part 2 checkpoint (epoch 80) reaches FID 44.6105. This is an absolute improvement of 17.7066 (approximately 28.4% relative reduction), indicating that continued optimization improves alignment with the WikiArt distribution in Inception feature space.

**FID vs IS behavior.** IS varies non-monotonically across checkpoints, which is common for multi-modal datasets. In contrast, FID provides a clearer signal and decreases steadily after resuming training, making it more reliable for checkpoint selection in this setting.

**Qualitative consistency.** Sample grids show improved texture fidelity and more coherent global structure at later epochs, consistent with the quantitative FID trend. Overall, selecting checkpoints by FID and corroborating with qualitative samples yields a robust evaluation.

## References

- [1] A. Radford, L. Metz, and S. Chintala. Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks. ICLR, 2016.
- [2] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. NeurIPS, 2017.
- [3] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna. Rethinking the Inception Architecture for Computer Vision. CVPR, 2016.